

Justin Paneto

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Summary

Strategic and technically rigorous **Senior Data Engineer** with 10+ years of experience designing, building, and optimizing scalable data platforms across **SaaS**, **Finance**, and **Government** domains. Expert in **Python**, **Spark**, and **SQL**, with deep experience in **streaming architecture**, **data lakehouse design**, and **ML-ready pipelines**. Adept at orchestrating workflows with **Airflow**, deploying cloud-native infrastructure on **AWS**, **GCP**, and **Azure**, and enabling analytics, reporting, and AI/ML use cases. Known for mentoring engineers, leading architecture reviews, and delivering resilient, high-throughput data systems.

Skills

- **Languages:** Python 3.11+, SQL, Scala, Bash
- **Data Engineering:** Apache Spark, PySpark, Pandas, Dask, Airflow, dbt, Kafka, Flink, Beam
- **Storage & Modeling:** Delta Lake, Iceberg, Parquet, ORC, Avro, Data Vault, Star Schema, Snowflake
- **Databases:** PostgreSQL, MySQL, BigQuery, Redshift, Snowflake, MongoDB
- **Cloud Platforms:** AWS (Glue, EMR, S3, Lambda, RDS), GCP (Dataflow, BigQuery, Pub/Sub), Azure (Data Factory, Synapse, Blob Storage)
- **Orchestration & Workflow:** Apache Airflow, Prefect, Dagster, Luigi
- **Streaming & Messaging:** Kafka, Kinesis, Pub/Sub, Spark Structured Streaming
- **DevOps & CI/CD:** Docker, Kubernetes, Terraform, GitHub Actions, Jenkins
- **Security & Governance:** IAM, RBAC, Data Lineage, Encryption at Rest/In Transit, GDPR, HIPAA
- **Monitoring & Observability:** Prometheus, Grafana, ELK Stack, CloudWatch, OpenTelemetry
- **AI/ML Integration:** Feature Stores, MLflow, TensorFlow, scikit-learn, SageMaker Pipelines
- **Soft Skills:** Technical Leadership, Mentorship, Agile Delivery, Documentation

Professional Experience

Senior Data Engineer | Accenture

Chicago, IL | July 2023 – Present

- Architected distributed data pipelines using **PySpark**, **Airflow**, and **Delta Lake**, processing 10TB+ daily across **AWS Glue**, **EMR**, and **S3**.
- Designed real-time streaming architecture using **Kafka**, **Spark Structured Streaming**, and **Kinesis**, enabling sub-second ingestion for compliance and fraud detection.
- Built ML-ready feature pipelines using **dbt**, **Pandas**, and **MLflow**, supporting model training, versioning, and online/offline parity.
- Implemented data lakehouse architecture using **Apache Iceberg**, **Parquet**, and **Hive Metastore**, enabling ACID transactions and time travel.
- Orchestrated ETL workflows with **Airflow**, implementing dynamic DAGs, SLA monitoring, and retry logic across environments.

- Developed data quality frameworks using Great Expectations, custom validators, and alerting via Prometheus/Grafana.
- Designed dimensional models and fact tables using **Star Schema** and **Data Vault**, enabling scalable BI and analytics.
- Integrated third-party APIs and external data sources using **Python**, **OAuth2**, and **batch ingestion pipelines**.
- Automated CI/CD pipelines using **GitHub Actions**, **Docker**, and **Terraform**, deploying data services to **EKS** and **Lambda**.
- Implemented fine-grained access control using **AWS IAM**, **Lake Formation**, and **column-level encryption**.
- Conducted performance tuning using **Spark UI**, **query plan analysis**, and **partition pruning**, reducing job runtimes by 60%.
- Led architecture reviews, authored RFCs, and drove adoption of **data mesh principles**, **event-driven ingestion**, and **schema evolution strategies**.
- Mentored junior engineers in Spark optimization, Airflow DAG design, and cloud-native data engineering.
- Delivered secure government-grade pipelines with **audit logging**, **data masking**, and **PII redaction**.
- Authored onboarding guides, lineage diagrams, and internal documentation portals using **MkDocs**, **dbt Docs**, and **Confluence**.

Senior Data Engineer | Google

Mountain View, CA / Jan 2018 – Jun 2023

- Led development of scalable data platforms using **Apache Beam**, **Dataflow**, and **BigQuery**, supporting real-time analytics and ML workloads.
- Built streaming ingestion pipelines using **Pub/Sub**, **Kafka**, and **Beam**, enabling low-latency data delivery for internal dashboards and alerting systems.
- Designed and maintained data marts using **dbt**, **BigQuery**, and **Looker**, supporting self-service analytics and KPI reporting.
- Developed ETL pipelines using **Airflow**, **Python**, and **Cloud Functions**, integrating internal APIs, third-party sources, and legacy systems.
- Implemented data governance frameworks using **Data Catalog**, **IAM policies**, and **sensitive data classification**.
- Created feature stores and ML pipelines using **Vertex AI**, **TensorFlow**, and **MLflow**, supporting experimentation and deployment.
- Built internal tools for data profiling, schema validation, and lineage tracking using **Streamlit**, **Flask**, and **OpenLineage**.
- Automated infrastructure provisioning using **Terraform**, deploying data services across **GKE**, **Cloud Run**, and **Cloud Composer**.
- Conducted performance profiling using **Dataflow logs**, **BigQuery optimizer**, and **Spark UI**, improving throughput and cost efficiency.
- Led cross-functional architecture reviews, authored technical design documents, and contributed to platform-wide data strategy.
- Mentored engineers across teams, conducted pair programming sessions, and led workshops on **streaming architecture**, **data modeling**, and **ML Ops**.
- Delivered secure modules for education and government clients, implementing **data masking**, **audit trails**, and **compliance logging**.

- Created automated regression tests using **pytest**, **Great Expectations**, and **Postman**, integrated into CI pipelines.
- Authored internal documentation portals using **Markdown**, **Sphinx**, and **GitHub Wiki**, including onboarding guides and data contracts.

Data Engineer | Intuit

Mountain View, CA | May 2015 – Dec 2017

- Developed financial ETL pipelines using **Python**, **SQLAlchemy**, and **Airflow**, supporting tax, payroll, and compliance workflows.
- Built batch ingestion systems using **Flask**, **cron jobs**, and **PostgreSQL**, integrating third-party APIs and legacy data sources.
- Designed and optimized schemas for **SQL Server** and **Redshift**, applying indexing, partitioning, and stored procedures.
- Created data validation and reconciliation tools using **pandas**, **Jupyter**, and **custom rule engines**, improving data accuracy and auditability.
- Delivered secure modules with **RBAC**, **data encryption**, and audit logging, aligned with IRS and SOC2 compliance standards.
- Refactored legacy ETL scripts into modular, testable components using **PEP8**, **type hints**, and **unit tests**.
- Automated deployment workflows using **TeamCity**, **Docker**, and **Fabric**, improving release velocity and rollback reliability.
- Built internal dashboards for compliance tracking and operational reporting using **Dash**, **Plotly**, and **SSRS**.
- Participated in Agile ceremonies, contributed to sprint planning, and collaborated with QA and product teams on release readiness.
- Authored technical documentation, onboarding guides, and internal wiki pages using **Markdown** and **Confluence**.

Education

Montclair State University

📍 Montclair, NJ

🎓 Bachelor of Science in Computer Science

📅 Sep 2011 – Apr 2015